

Connor Savugot

B.S. in Computer Engineering | North Carolina State University | Graduated May 2026

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SKILLS

Programming & Data: Embedded C, C++, Python, MATLAB, Bash, SQL, VBA, Microsoft Access, Excel

Embedded Systems: STM32G474, dsPIC33CK, PIC32, TI C2000/TMS320, MSP430FR2355, TI AM62; bare-metal firmware, FreeRTOS, Embedded Linux, Yocto/Arago

Interfaces: CAN 2.0, CAN FD, J1939, I2C, SPI, UART/RS-232, ADC, PWM

Hardware & Debugging: KiCad schematics; PCB review in KiCad, P-CAD, and Altium; JTAG, oscilloscopes, logic/protocol analyzers; Git/GitLab

EXPERIENCE

AEGIS Power Systems | Embedded Firmware Engineer

May 2026–Present

- ▷ Write and debug dsPIC33CK firmware for embedded power hardware, including CAN, ADC, PWM, I2C, SPI, UART, and timers; verify electrical signals and bus traffic with JTAG, oscilloscopes, and logic analyzers.
- ▷ Use schematics and PCB layouts from KiCad, P-CAD, and Altium to trace system failures across wiring, power, firmware, CAN traffic, device settings, Linux services, and application software.
- ▷ Integrate CAN-connected power devices with a TI AM62 Embedded Linux controller, including Yocto/Arago images, Linux kernels, device trees, boot troubleshooting, and an external-watchdog service.

AEGIS Power Systems | Embedded Systems Intern | Computer Science Intern

Summers 2024 & 2025

Selected work across both internships

- ▷ Wrote bare-metal C firmware for a TMS320-controlled bidirectional power converter with 12 PWM A/B pairs (24 outputs); configured frequency, duty cycle, and phase for each A output and set its paired B output to follow or invert it.
- ▷ Worked on a two-processor power system in which a FreeRTOS PIC32 handled I2C, SPI, CAN/J1939, UART, and firmware updates while a bare-metal TMS320 ran the real-time converter controls.
- ▷ Built a firmware update and recovery workflow that transferred checked image chunks over TCP, stored new and golden images in PIC32 SPI flash, and programmed and verified the TMS320 over UART.

Production and engineering tools

- ▷ Simplified TI's C++ serial flasher for the C2000 TMS320F28379D by removing unused device, CPU2, and dual-boot options, allowing production staff to program precompiled firmware without choosing engineering-only settings.
- ▷ Wrote instructions for programming production units and building release images in CCS; also built internal device-discovery, programming, and diagnostic tools for engineers and production staff.

TEAM Industries | Engineering Intern

Summers 2021 & 2022

- ▷ Built Excel/VBA tools that parsed inspection data and flagged parts outside dimensional and spline tolerances.
- ▷ Created and maintained a 17,000+ entry tooling where-used list for the Inrite plant, linking each tool to part numbers, operation numbers, storage locations, and vendors with custom lookup formulas and data-validation rules.

PROJECTS

Manufacturing Test Automation | SQL, VBA, Access, Excel

TEAM Industries

- ▷ Built tools for an Acceptance Test Procedure database that parsed and checked test results, retrieved earlier records, and entered new results automatically.

EV Active Sensor Adapter | Battery Monitoring & CAN

NC State University

- ▷ Wrote STM32G474 firmware for a custom EV sensor board that read ten thermistors through the ADC, communicated with a BQ79616 battery monitor over UART, and sent measurements over extended-ID CAN.
- ▷ Implemented the BQ79616 wake and setup sequence, message framing, CRC checks, cell-voltage reads, and fault reporting; documented board bring-up and troubleshooting for the project team.

C++ Microarchitecture Simulators

NC State University

- ▷ Built configurable cache, branch-predictor, and out-of-order pipeline simulators in C++; processed instruction traces and measured cache misses, branch accuracy, memory traffic, and instructions per cycle.

TEACHING EXPERIENCE

ECE 306 Embedded Systems | Teaching Assistant & Course Mentor

Spring 2025–Spring 2026

- ▷ Taught supplemental lessons and helped students debug MSP430FR2355 firmware and custom PCBs across three semesters, including register, timing, peripheral, toolchain, wiring, and soldering problems.